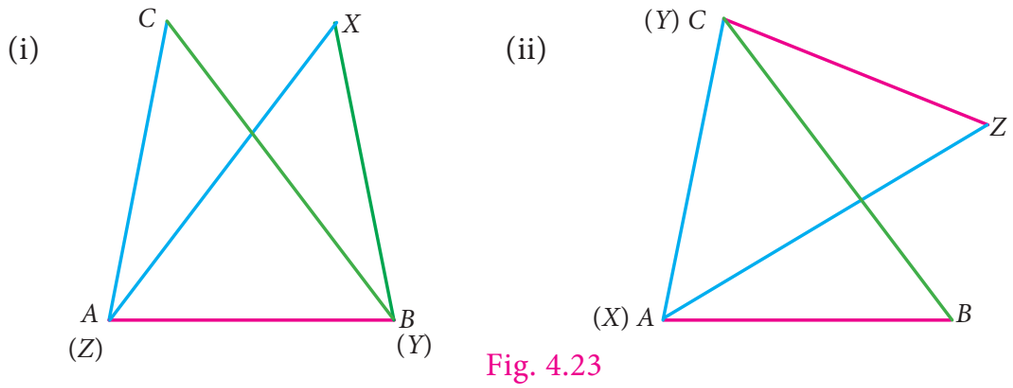


Fig. 4.23

This does not imply that the two triangles are not congruent.

So to confirm the congruency of given two triangles, we have to check the congruency of corresponding sides and corresponding angles.

Hence, the above triangles $\triangle ABC \cong \triangle XYZ$ are congruent.

So, we conclude that, if all the sides and all the angles of one triangle are equal to the corresponding sides and angles of another triangle, then the two triangles are congruent to each other.

4.4.6 Conditions for Triangles to be Congruent

We learnt to check the congruency of triangles by superposition method. Let us check the congruency of triangles using appropriate measures which will be very useful. We can study them as criterions to check the congruency of triangles.

Look at the following triangle (Fig. 4.24).

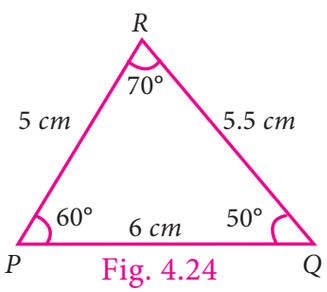


Fig. 4.24

To draw a triangle congruent to the given triangle do we need to know all measures of sides and angles of the triangle?

To construct a triangle, we need only three measures. Those three measures can be any of the following.

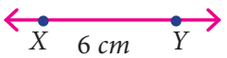
1. The lengths of all three sides. (or)
2. The lengths of two sides and the angle included between those two sides. (or)
3. Two angles and the length of the side included by angles.

We can try the above conditions one by one.

1. Side–Side–Side congruence criterion (SSS) - The lengths of all three sides are given

Draw a triangle XYZ given that $XY = 6\ cm$, $YZ = 5.5\ cm$ and $ZX = 5\ cm$

Step 1: Draw a line. Mark X and Y on the line such that $XY = 6\ cm$



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